



electronic file preparation

(Rev. 09/04/02)

accepted electronic media

- CD
- Floppy
- zip (250 mb & below)
- jazz (2gb & below)

accepted software

Mac

quark 4.1
pagemaker 7.0
adobe illustrator 9.0**
adobe photoshop 6.0
freehand 9.0**
indesign 2.0

PC

quark 4.1
pagemaker 7.0
adobe illustrator 9.0**
adobe photoshop 6.0
corel draw 10.0**
publisher 2002*
microsoft word*

* We can accept MS Word and Publisher 2002 for electronic output but due to deficiencies in these inexpensive programs we cannot guarantee final results. The most common problems are uncontrollable color variations, font flow and type running up and downhill. We highly recommend using quark express for critical layouts. **Illustration and photo programs (adobe illustrator, photoshop, corel draw, freehand, etc.) should be used for illustrations and photos only. They should not be used for setting up entire layouts.

Electronic File Transfer

E-mail

We can accept your electronic file via e-mail. Your customer service representative will assign a personal e-mail address to you if you wish. You may also e-mail your file to our composition department at prepress@dasprint.com

Electronic files must include all necessary fonts and graphics saved within one folder and compressed using a program such as Stuffit (MAC) or WinZip (PC).

Files should be e-mailed as attachments along with detailed instructions in the message box. It is also recommended that you attach a PDF file of the entire document.

FTP

We can accept your file via FTP transfer, please see the instructions on our website for further information. FTP

File Corrections

Files that do not meet David A. Smith specifications will be returned immediately to the customer or, if possible, corrected on site. Poorly supplied files will delay production and may jeopardize target due dates. For further information on electronic file preparation please see additional tip information or contact our composition department.

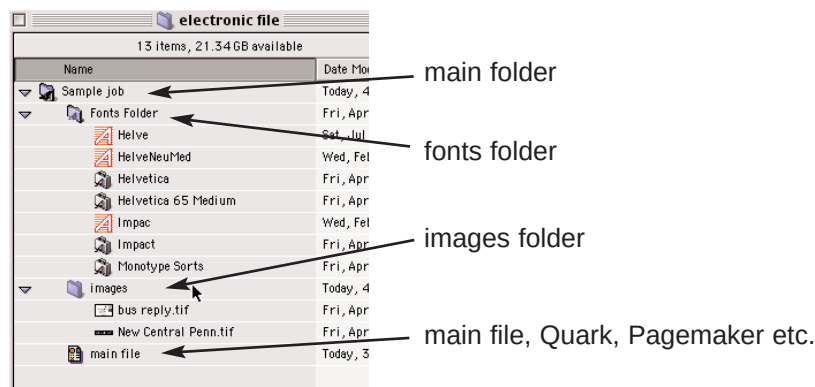
Additional Charges

David A. Smith, Inc. reserves the right to impose additional charges for any time intensive labor necessary for correcting a poorly supplied file. These charges will be incurred based on an hourly rate for any file deemed poor in nature that requires manipulation. Some of the most recurring examples that would qualify under this topic would include: converting a file from RGB to CMYK, reflowing of text, exuberant type changes and font issues

Electronic File Guidelines

Disk Preparation

- All electronic files should be organized in the following manner:



Main File

All supplied electronic files should include only those intended for output.

(Quark, PageMaker, etc.) All files should be set up to the exact page size necessary for printing. Do not make your document oversized and/or add your own crop marks. For example, if the document has a trim size of 8 1/2" x 11", then the page size of your document should be formatted the same. Also, please do not set up your own printer spreads.

Fonts

Copies of all screen and printer fonts used must be supplied for all pages and EPS files. Avoid using true type fonts. These fonts have a tendency of converting to PostScript Bitmap when output through high-end devices. This conversion will most likely result in the reflowing of text and an undesirable bold and/or italic appearance. David A. Smith Printing, Inc. will not be responsible for any reflow or font substitution occurred in files provided with true type fonts. In order to limit confusion, DAS strongly suggests that the customer supply only those fonts essential for the printed piece.

Graphics

All graphics used in job file need to be supplied in the proper resolution.

Halftones= 300 dpi

Linear/Bitmap images= 1200 dpi

Graphics created in drawing or photo software (Illustrator, Photoshop) should be saved as an EPS or TIFF file and be included in the supplied electronic file. Some programs such as PageMaker and Corel Draw will allow you to embed your graphics. We require that all graphics be linked, not embedded. In addition, we advise furnishing a full resolution copy of every linked graphic used in your publication.

Colors

One of the most common problems with electronic files is color management. If the supplied job is to print in 4/Color process, then all graphics must be formatted in CMYK, not RGB. Graphics intended to print as PMS matches should be set up for process separations if applicable (see Illustrator section). If the job is to print in black and PMS 280, then all images to print in the second color must be set to that corresponding color. Any colors that are not used from the color palette should be deleted.

-See Color Management for further clarification

Laser Dummy

All disk files submitted must include a laser dummy. If a job requires any special folds or back ups, then they should be annotated on the dummy. It is important to output the dummy to final page size. If it is necessary to deviate from the original size then show then show the enlargement or reduction on the proof. The supplied dummy should be clean and unmarked without revisions. The dummy will be a reference tool for various departments throughout production. If the file is going to be sent electronically via e-mail, an updated PDF file should be included to enable us to run our own copy. In some circumstances, it is suggested to run color separated lasers to show color breaks.

Trapping

Do not change the trapping preferences in your document. Our Katana RIP will automatically trap your job. Trapping preferences that are changed could affect the final output.

PDF files (this option should only be used by designers with adequate knowledge on writing PDF files)

Our new Katana RIP allows us to separate PDF files for 4/C and spot color printing. However, because we cannot alter a customer's PDF file certain settings should be applied in the distiller before writing the PDF file.

The native file (Quark, PageMaker etc.) should be set up in the correct colors. If the job is to print 2/color and the 2nd color in the file is being written as a RGB or CMYK build, it will not color separate correctly. If the job is to print PMS 280 and black, the electronic file should be set up as PMS 280 and black.

All jobs to be printed in 4/C process and any color halftones are to be saved in CMYK mode before writing the PDF file. RGB files are notorious for causing problems.

Postscript Settings (the following settings should be used for saving hi-res .ps files)

- Printer Description: Acrobat Distiller
- Print Colors: Composite CMYK (process & pms spot work)
Composite Grayscale (black & white only)
- Bleed: .125" should be specified for bleed amount
- Size: should be set to final trim size
- Full Resolution Tiff: "needs to be checked on"

These settings can be saved as a print setting in most applications.

Distiller Settings (The following distiller settings should be changed under job options or preferences before distilling a file for high-resolution output)

- Resolution: 2400
- Page Size: set to final trim size
- Compression:
 - Color Images: bicubic down sampling should be set to 300 dpi
 - Grayscale Images: bicubic down sampling should be set to 300 dpi
 - Monochrome Images: bicubic down sampling should be set to 1200 dpi
- Fonts: embed all fonts setting should be on

These settings can be saved as a ticket that can be used whenever needed.

*This ticket should only be used for high-resolution requirements, as these settings will greatly increase the final size. Once this ticket has been saved with the proper setting it can be used again without the need to readjust.

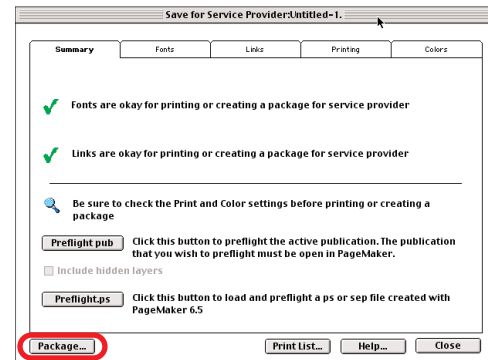
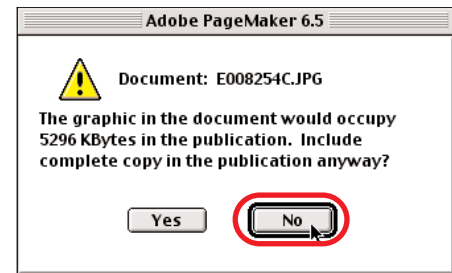
PageMaker

Graphics

Do not embed your graphics or photos! We require that all graphics be linked, not embedded. In order to insure an accurate reproduction, a full resolution copy of every linked graphic is also requested.

PageMaker will prompt you when the graphic is placed: "The graphic in the document would occupy XXX Kbytes in the publication. Include complete copy in the publication anyway?" select NO.

If YES is selected, the entire graphic is embedded within the document. This can lead to ripping problems as well as make the PageMaker file itself extremely large. The larger file will prolong the time it takes to print, scroll pages, etc...



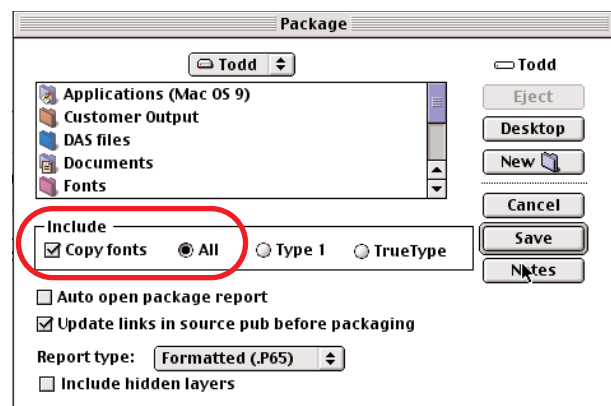
Color

Always check for color inconsistencies. When printing 2 or 3 color jobs you should only have one PMS definition per color. If there is more than one, it means there will be unnecessary plates generated. A good way to avoid problems with color is to set up any graphics that are going to use PMS colors beforehand. Then import them into your layout program and use those color definitions to color text. Any unused colors should be deleted from the color palette to limit confusion.

'Save For Service Provider'

PageMaker has a plug in tool called 'Save For Service Provider' located under utilities/plugins. This plug-in will collect all fonts and graphics that were used in the file and nest them into one folder.

- Produce a new folder and place it in a location different from that of the original PageMaker file.
- Make sure 'copy all fonts' is selected.
- Start the 'Save For Service Provider' program and direct it to gather all files in the folder you created.
- This utility should be used for collecting all jobs.



Quark

Color

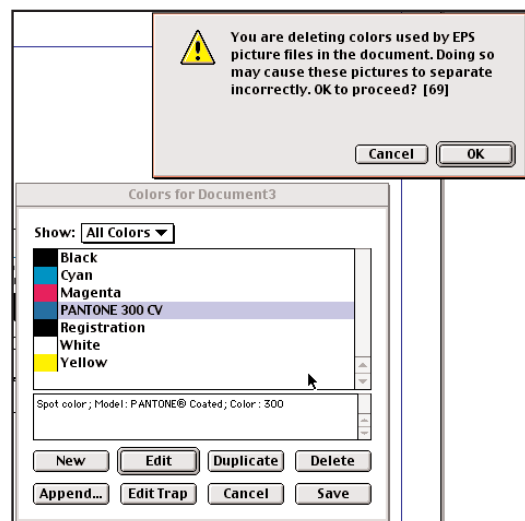
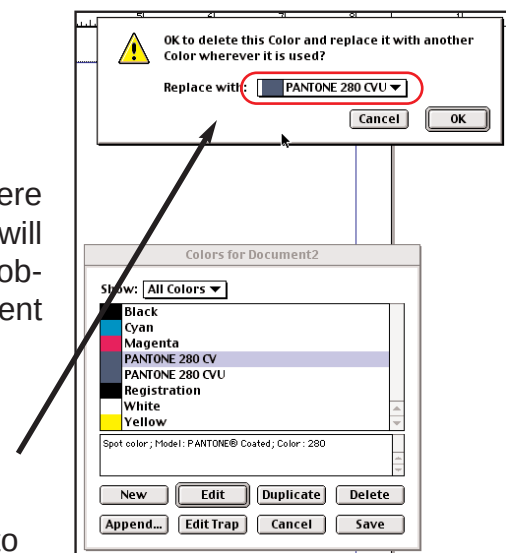
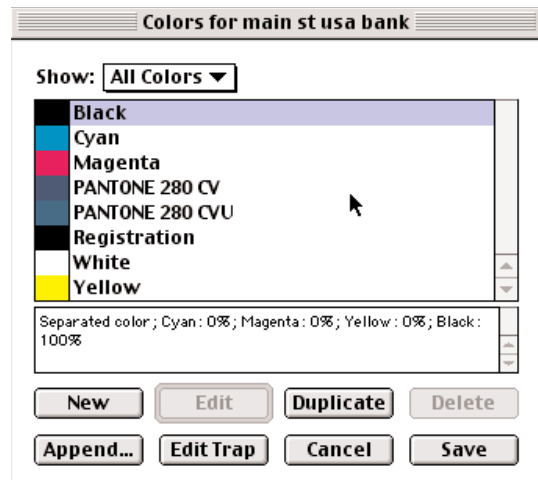
Always check for color inconsistencies. When printing 2 or 3 color jobs you should only have one PMS definition per color. If there is more than one, it means there will be unnecessary plates generated. A good way to avoid problems with color is to set up any graphics that are going to use PMS colors beforehand. Then import them into your layout program and use those color definitions to color text. Any unused colors should be deleted from the color palette to avoid confusion.

This color palette is showing a common problem. There are too many definitions for PMS 280 and this will cause two plates to print. This is a very common problem caused by importing graphics that have a different color tag than those set up in Quark.

Solutions

If you are importing several graphics from different programs using the same PMS color you will have to edit the specific color. Do this by naming the PMS color in one of the programs so that it matches the name in the other exactly. If you are importing just one graphic and are getting the extra color tag from Quark, you can delete the Quark color and tell the program to use the imported tag to color the rest of the document.

If you see this error message when trying to replace colors, cancel immediately. This means you are trying to delete a color used in an imported graphic.



Photoshop

Halftones

Halftones should be scanned @ 300 dpi/line art should be scanned @ 1200 dpi.

RGB

RGB stands for red, green and blue. These colors comprise of those used on computer monitors and web images. They are not suitable for any applications in offset printing. All RGB images must be converted to CMYK or black before being sent for output. Color shifts may occur when converting RGB images to CMYK and may be necessary when performing color corrections through Photoshop or other photo editing software. Please consult your owner's manual for more details on color corrections. Be sure to keep a copy of the original RGB file if it might be needed for web use at a later date.

Type

Type can be used in a Photoshop image. However, because of the fact that it is converted to a bitmap format, the edges will appear slightly bitmapped. (See image at right)



Web Images

It is very tempting to copy that great image on the web and paste it into your document. Don't Do It!!! Web images are generally saved in 72 dpi, which looks good on a computer screen (because computer screens are viewed at 72 dpi) but will look bitmapped when offset printed.

Image Solutions

Many high-resolution (300 dpi) images can be found and downloaded on the web through stock services. Photo Disc, EyeWire and Corbis are some of the more popular and reliable. Costs can range from \$30 - \$175 depending on the image size and resolution needed. CD collections that contain hundreds of images can also be purchased from these companies. A lot of our clients choose this option for building high-resolution, image libraries because it greatly reduces the cost per image.

Changing Resolution

You cannot take a 72 dpi image and just change the resolution to 300 dpi in photo editing software and expect to get better results. When you start with a low-resolution image you cannot add detail. Changing an image from 72 dpi to 300 dpi is in effect only adding additional pixels, not enhancing it. You must start with quality (300 dpi) images in order to get like results.



image downloaded from the web 72 dpi



same image converted to 300 dpi

Good Photography

There is no substitution for good photography. The best equipment, knowledgeable operators and everything saved to the proper resolutions cannot replace a quality photo. If you have poor photographs, the end result will be poor.

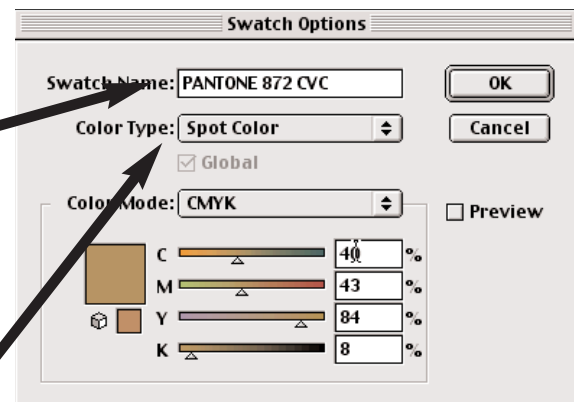
Illustrator/Freehand

Colors

Depending on what application you are using, the same pantone color may be identified under different tags. Different applications identify the same pantone color with different tags. We find the best way to set up two or three color jobs with imported graphics is to get the color the way we want it in the illustration program first. This will insure that the illustration file is being imported into Quark and/or PageMaker along with the illustration to set up color in our main document.

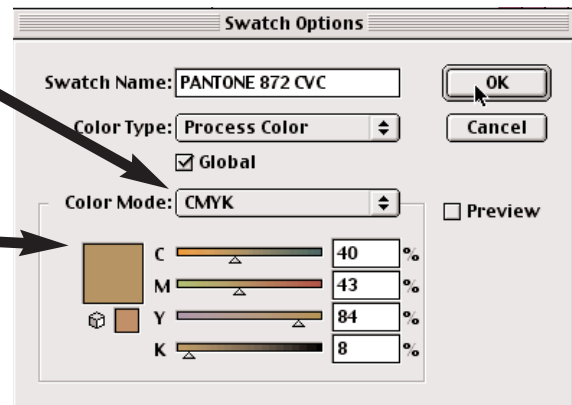
Color Definition

The color defined here will import along with the illustration file into Quark and/or PageMaker. Set up the proper color definitions here and they can be reused in Quark and/or PageMaker. This will help to avoid multiple color definitions.



Spot Color: Be sure to check whether or not files are to print as spot colors.

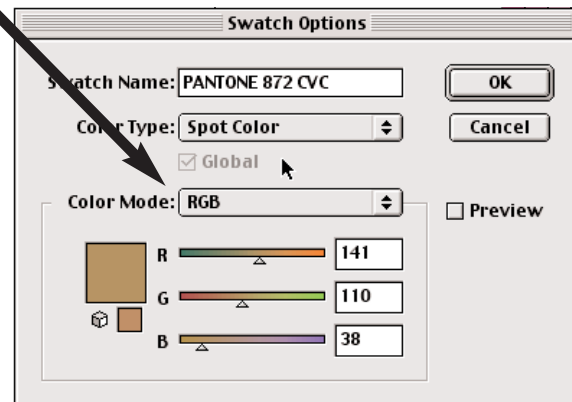
CMYK: If the file is to print in CMYK, this should be confirmed. CMYK percentages should NEVER be altered if you are trying to match a specific pms color for your final printed piece.



RGB: Clip art and graphics created in illustration software should never be saved in RGB format for offset printing.

Fonts

- It is imperative that any font used in an illustration program be included on the final disk. A good way to avoid potential problems is to convert all of your fonts to outlines or paths. (Illustrator: Type Menu/create outlines) Remember to keep a copy of your original file with the fonts intact because once you convert to outline, you can't go back.



(Graphics shown above are from Adobe Illustrator. Freehand has similar commands.)

Images

Halftones/photos: halftones need to be scanned @ 300 dpi for optimal print quality.

Web images: images downloaded from the web are generally 72 dpi and will not yield acceptable results in offset printing. (See below)



graphic @ 300dpi



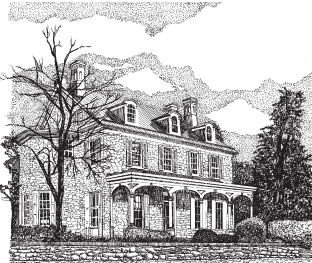
graphic @ 150dpi



graphic @ 72dpi

All images should be saved in the appropriate mode. Images intended to print 4/C process should be saved as CMYK, not RGB. Images intended to print as black or one color should be saved as grayscale or with the proper PMS color specification. Duotones should be saved in duotone mode in Photoshop.

Line art/Bitmaps: should be scanned @ 1200 dpi for optimal results.



graphic @ 1200dpi



graphic @ 300dpi



graphic @ 100dpi

Linked images: Some layout programs will allow you to place/embed images into the document without a link. This may present problems with the file size and outputting to film. David A. Smith Printing, Inc. requires all images to be linked and be included in an images folder. (See Electronic File Guidelines)

Digital Cameras

The quality of digital camera images varies greatly depending on the model you choose. Digital cameras capture the image and divide it into pixels. A pixel is a small block of digital information. A camera with a high pixel count (5 megapixels-5 million pixels) will create a better image than a camera with a low pixel count (1 megapixel-1 million pixels). A 5-mega-pixel camera will capture each image and divide it into 5 million small blocks. This will create a sharp and detailed image. Cameras with a low pixel count do not have the ability to create a high quality image for printing or developing. They do however produce an image sufficient for posting on the Internet. Most images on the Internet are at a resolution of 72 dpi (dots per inch). A 72 dpi image is small enough to load easily onto a web page and detailed enough to be recognizable on a computer monitor. These images may appear blurry and would lack detail if used for offset printing. Digital cameras take the 72 dpi image and create a large file such as 22" X 16.67". If you use the image in a document and reduce it to 25% then you will have 288 dpi. When you reduce an image to 25% it is like shrinking it on a photocopier. You will have four times as many dots as in the original area. The image will appear sharper because it is reduced but actually there is no increased detail. An image shot at a certain setting it can never be improved. Saving a 72dpi image at 300 dpi merely divides a blurry image into more dots. It does not improve the photo. Low pixel images will more than likely have a bit of a blurry or block-like appearance when printed or developed.

All digital cameras save the images as RGB (Red, Green and Blue). Red, green and blue represent the three colors of light that are used to capture an image. Televisions and computer monitors use these three colors of light to make all other colors. If you look closely at your television or monitor you can see the RGB blocks. Printing uses four ink colors to reproduce your photographs. This is called 4-color process and uses layers of Cyan, Magenta, Yellow and Black inks to recreate the image. All RGB images must be converted to either Black or CMYK for printing using offset methods.

Most digital cameras only allow the image to be created as a JPEG (Joint Photographers Expert Group). A JPEG is considered a lossy format. That means the camera strips away much of the digital information from what the camera actually captured. A JPEG chooses blocks from your image and discards them then saves the remaining blocks to your memory storage device. This is done to make the file size smaller. A JPEG saves the image at a resolution of 72 dpi. Some high-end cameras will allow you to take pictures as a TIFF (Tagged Image File Format). A TIFF is a non-lossy format that will produce an excellent image. A TIFF does not discard any of the blocks so the entire image is saved. A TIFF saves the image at a resolution of 300 dpi creating a much cleaner image. These files are large and may approach 14MB in size per photo.

The next variable for digital photography is memory. Memory is the amount of information you can hold on a removable storage device. These devices are sometimes 3 1/2" floppy disks but are more than likely to be a Smart Media or Flash Card about the size of a matchbook. These cards are available in memory capacities of 8MB (8 megabytes) up to 256MB. The memory on these cards is a fixed amount. You might be able to take thirty-two lowest resolution images on an 8MB card or you might only be able to take two highest resolution images on the same card. Many people make the mistake of trying to fit too many pictures on their memory card. They change the setting on their camera to the lowest resolution and take as many pictures as possible until the memory card is full. This means that each picture taken will have a small amount of digital information for the camera to produce the image. The result will be low quality photos with poor color and detail.

When using a digital photo in your document be sure to link the original saved file. Do not send only a cropped image because cropping will further diminish the file size and image detail. You will probably want to show us how you wish the photo to be cropped. Simply create a second image of the file at the cropped size. We will then use your original saved file for the document and reduce it within the window to match your cropping. This will not reduce the file size so the printed piece will hold the detail of the original file.

Printing a digital photo with your \$100.00 ink jet printer at home or even the expensive photo printers available for \$400.00 may produce a satisfactory image. Both devices use liquid ink cartridges to produce the pictures. These images will always be inferior to prints made on actual photographic paper. Photographic paper uses layers of light sensitive emulsion rather than inks. Your local film processor should be able to make prints on photographic paper from your own CD. You will be pleasantly surprised by the results.

Fonts

Supplied Fonts

- All necessary fonts must be provided with every job. Customers should not provide us with every font in their library. Only enclose the fonts that are essential for each individual job. A lot of page layout programs provide 'collect for output' or 'collect for service provider' options and will collect all fonts and graphics used in a job and place them into a directory (Quark 5.0, PageMaker, Publisher). Please refer to the manual provided with your layout software for more detailed instructions. These functions should be utilized whenever possible. There are also third party utilities such as Flightcheck that can be purchased or downloaded that offer similar capabilities.

Q) Why can't a customer provide their entire library and DAS find and use fonts when necessary?

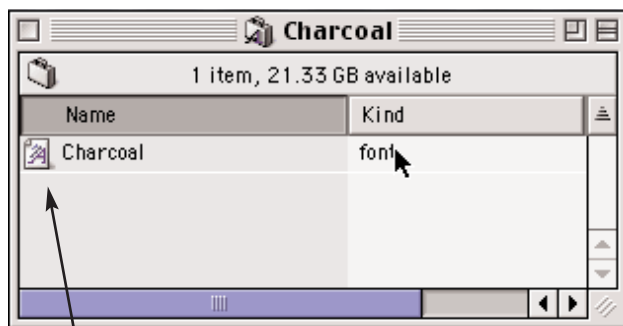
A) Because this is a very time consuming process that entails additional storing and time retrieving files from our own archive. It is the responsibility of the person preparing the file to provide all necessary items. David A. Smith Printing, Inc. reserves the right to charge for any additional time needed to correct poorly set files.

Avoid Using Menu Styles

Avoid applying menu styles to your fonts, such as bold, italic, outline, shadow, underline etc. There are no printer fonts for these files and the type will either default to black or not print at all. Solid type with a drop shadow can be created in the layout program using separate text boxes. Outline type should be created in the layout program and saved as paths/outlines. Always use the actual bold, or italics bold version of the font, not the menu bar.

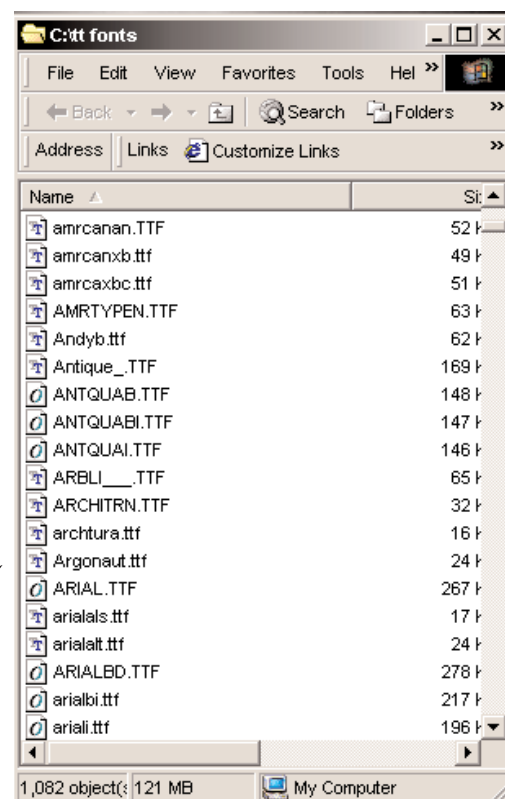
True type

We do not recommend using true type. These fonts have a tendency to convert to PostScript bitmap when output through high-end imagesetters. This can cause undesirable reflow and substitution issues.



true type font/mac
(notice 3 a's)

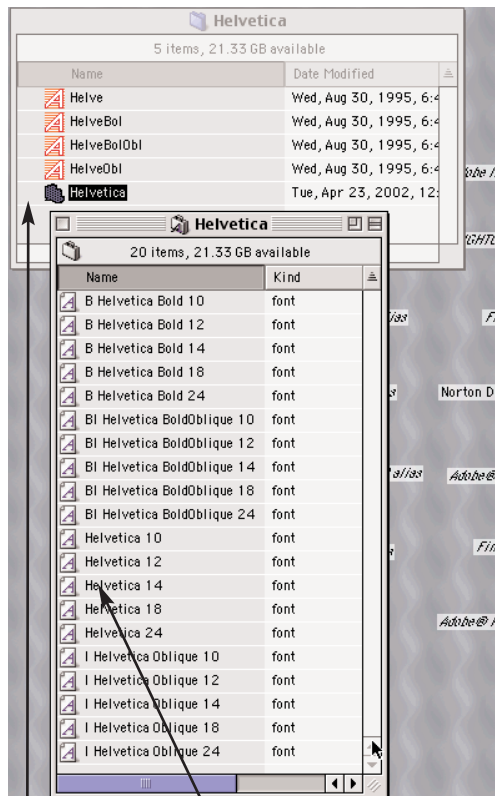
true type font/pc
(notice .TTF extension)



Type 1/PostScript

We recommend using Type 1/PostScript fonts whenever possible as these fonts are written in the PostScript language used by imagesetters. Type 1 fonts require two files for each font. These fonts are referred to as printer and screen. We will need both for each font.

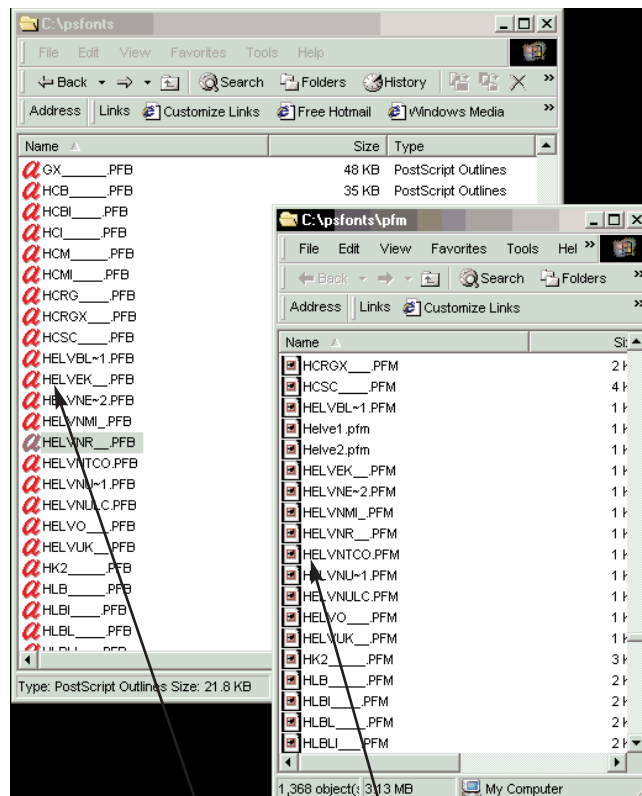
type 1 font/mac



printer font

screen font

type 1 font/pc



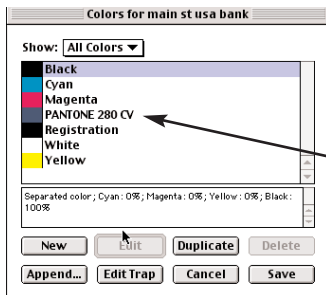
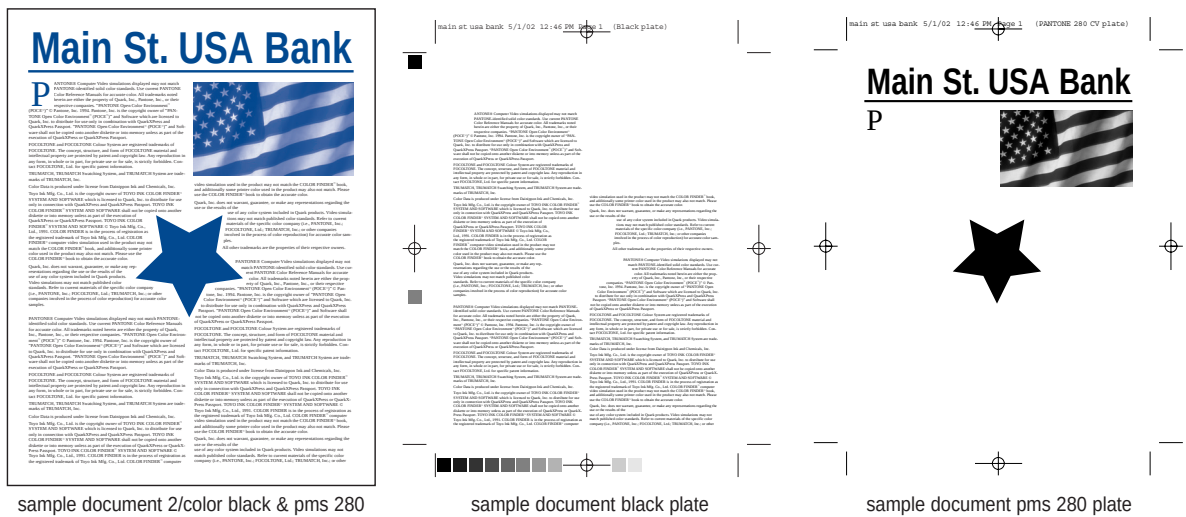
screen font

printer font

Important Note: Many files will print on laser printers because they emulate PostScript. This does not mean the files are suitable for high-end rips.

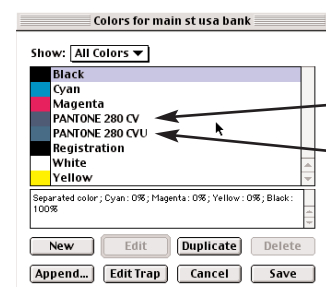
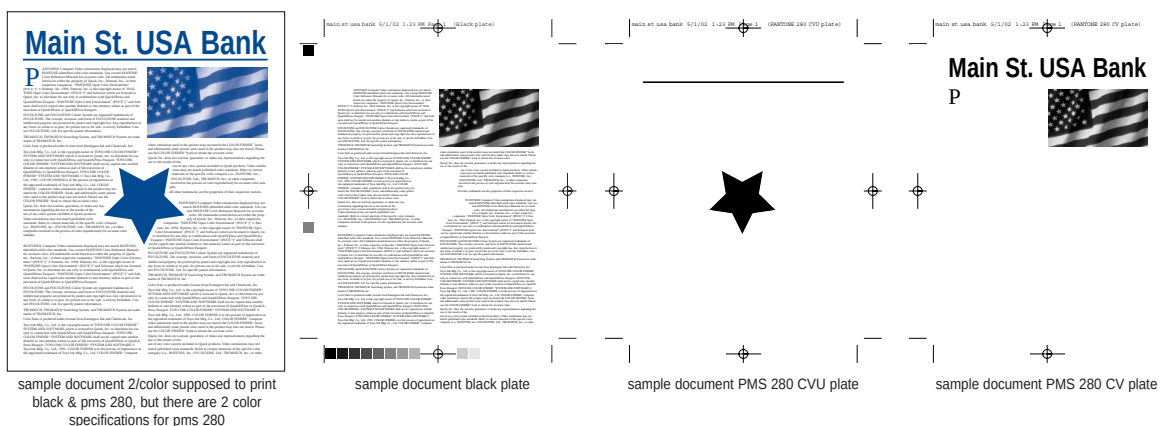
Color Management

PMS Jobs set up to print as spot colors should only include the necessary PMS colors. They should not include additional PMS color specifications.



This is how the color palette should look for the above document. One PMS 280 Color specification

Sample 2/c job with spot colors set up incorrectly:



This color palette is set up incorrectly. There should only be one PMS 280 color specification. The sample set up will output two plates for the same color when it should only need one. (See above)

Converting RGB to CMYK

Color shifts may occur when converting RGB images to CMYK. It may be necessary to perform your own color corrections using Photoshop or other photo editing software. Please refer to your owner's manual for more details pertaining to color corrections.



this image is CMYK

PMS Percentages

- When selecting a PMS color in Quark or PageMaker you want to match for use in 4/C printing. Allow the program to automatically calculate the CMYK percentages necessary to build that color. These percentages should never be changed. If they are, the color will not match its intended PMS color. You can also use this field to build custom CMYK colors for 4/C process printing. Use the name field to assign a generic name not to be confused with the PMS one.

